Gamified Real-Time sEMG Brain-Machine Interface

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SUMMARY

Spinal cord injury (SCI) is among the most common types of injuries in the U.S., affecting more than 300,000 people and at a rate of 18,000 new cases every year. Even though SCI disrupts the communication between the brain and the motor circuits located in the spinal cord, motor intent can still be detected reliably from residual muscle activity and could be used to control assitive devices.

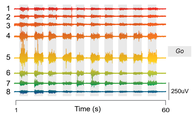
To this end, we developed an AI-based pipeline capable of decoding motor intent from surface electromyographic (sEMG) activity and translating it to play interactive games up to three degrees of freedom (DOF). We assessed in 5 able-bodied subjects the performance of the three different types of control modes: Discrete Control, Position Proportional Control, and Velocity Proportional Control to manipulate the gaming avatar's movement.

In these preliminary results, we found that velocity-proportional control outperformed position-proportional control and delivered comparable performance to discrete control in a gaming setup. Importantly, we could discriminate digits from wrist movement using only an 8-channel sEMG band placed on the forearm. Although preliminary, we believe that our results can be replicated in spinal cord injured population to enhance human-computer interaction.

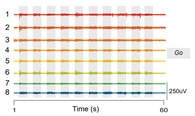
INTRODUCTION



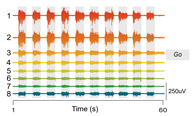
WRIST FLEXION



INDEX FLEXION



WRIST EXTENSION



METHODS

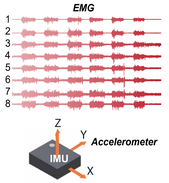
sEMG PLACEMENT





MINDROVE SPECS

- 8 Channel wireless sEMG device
- 8 + 2 Stainless steel dry electrodes
- EMG with 500 Hz sampling frequency
- IMU with 50 Hz sampling frequency



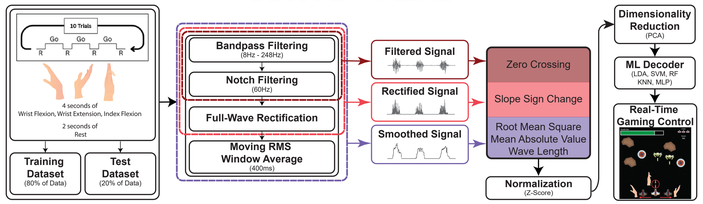
PARTICIPANTS



2 Male

3 Female

DECODER PIPELINE



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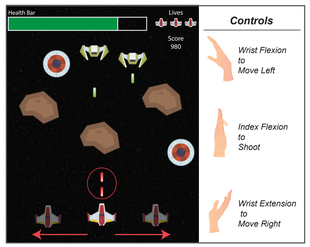
NML Website

RESULTS

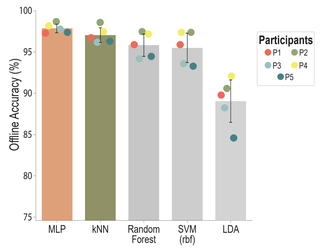
DISCRETE CONTROL

Participants achieved an average of 82.09% gaming performance with MLP based classifier

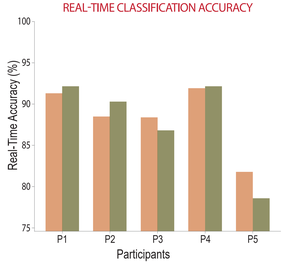
TASK



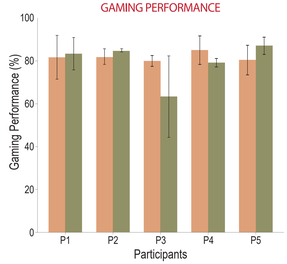
OFFLINE CLASSIFICATION ACCURACY



REAL-TIME CLASSIFICATION ACCURACY



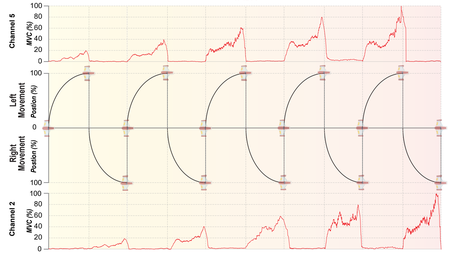
GAMING PERFORMANCE



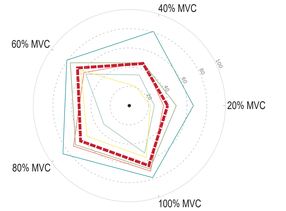
POSITION PROPORTIONAL CONTROL

Gaming performance decreased by 31.21% with the position-based controller compared to the discrete controller, reaching an average game performance of 50.87%

TASK



GAMING PERFORMANCE



EQUATION

$$i^* = \arg\max_i M_i^{(g)}$$

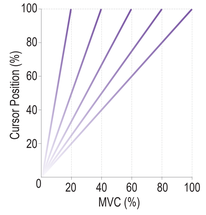
$$n = \min\left(\frac{p}{p_{\max}}, 1\right)$$

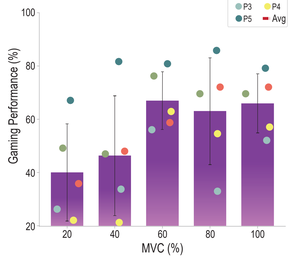
$$x_t^{(\text{left})} = x_c(1-n)$$

$$x_t^{(\text{right})} = x_c + n(W - x_c)$$

p : EMG value
$p_{\max}$: MVC
n : Normalized activation
x_c : Center (neutral) position
W : Right boundary of screen
$x_t^{(\text{left/right})}$: Target position
i : Channel index
$M_i^{(g)}$: MVC amplitude of channel i for gesture g
g : Gesture label
i^* : Selected (most active) channel

MVC VS POSITION OF CURSOR

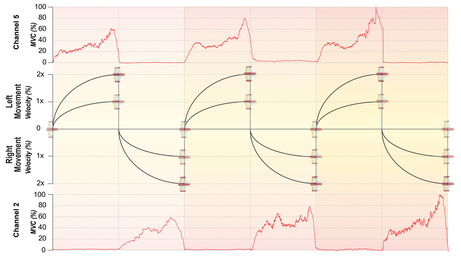




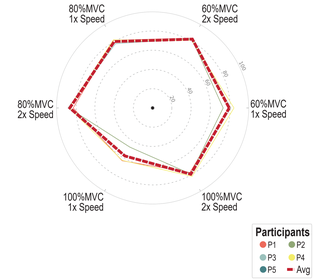
VELOCITY PROPORTIONAL CONTROL

Gaming performance improved by 35.48% with the velocity-based controller over the position-based controller, reaching an average game performance of 86.35%

TASK



GAMING PERFORMANCE



EQUATION

$$i^* = \arg\max_i M_i^{(g)}$$

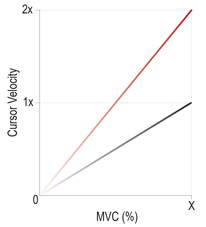
$$n = \min\left(\frac{p}{p_{\max}}, 1\right)$$

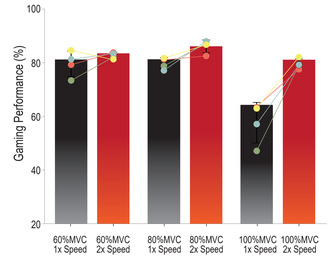
$$v_x^{(\text{left})} = -(V_0 + n)\,V_h$$

$$v_x^{(\text{right})} = +(V_0 + n)\,V_h$$

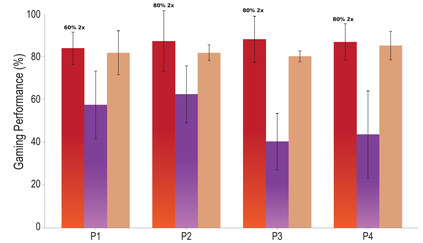
p : EMG value
$p_{\max}$: MVC
n : Normalized activation
V_0 : Velocity multiplier
V_h : Horizontal speed constant
$v_x^{(\text{left/right})}$: Horizontal velocity
i : Channel index
$M_i^{(g)}$: MVC amplitude of channel i for gesture g
g : Gesture label
i^* : Selected (most active) channel

MVC VS VELOCITY OF CURSOR

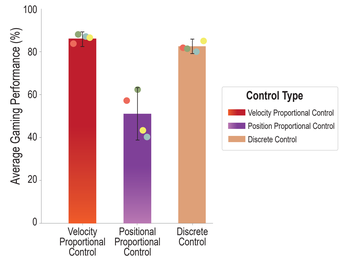




COMPARISON BETWEEN DIFFERENT TYPES OF CONTROL



AVERAGE PERFORMANCE



CONCLUSION

- This work demonstrates that AI-based decoding of motor intent from surface EMG enables intuitive game control with strong potential for neurorehabilitation.
- Velocity Proportional Control provided a more natural and engaging experience than Positional Proportional Control.
- Digit movements were decoded accurately using only eight EMG channels, and control was successfully extended to multi-degree-of-freedom (DOF) gameplay.
- Position based controller were affected by maxiumum muscle effort level.
- As a future direction, the system can be expanded by integrating Inertial Measurement Units (IMUs) with sEMG to capture both muscular effort and hand kinematics and further testing the use case in spinal cord injury (SCI) patients.